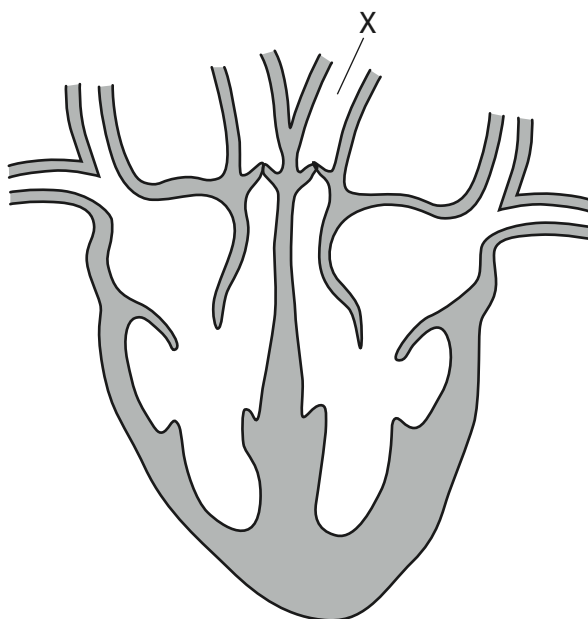


7. The diagram shows a section through a human heart.



(a) (i) Name blood vessel X.

1

(ii) Describe the evidence shown in the diagram that suggests the heart is in atrial systole.

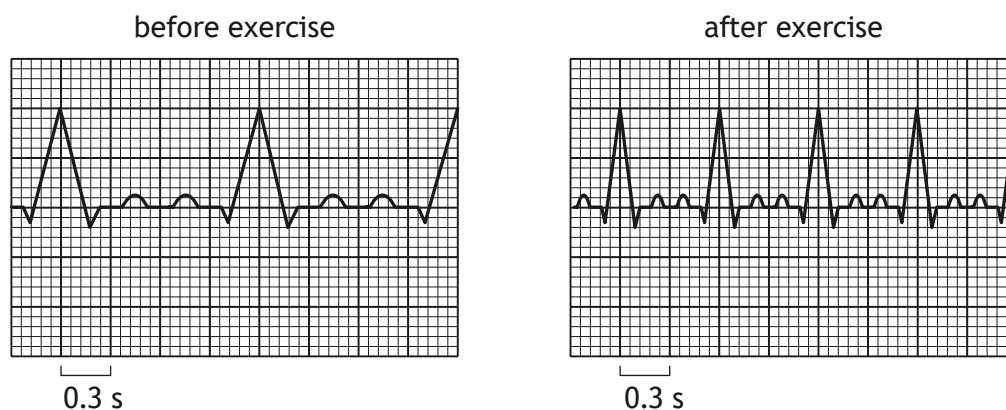
1

(b) Describe how impulses from the sino-atrial node (SAN) cause ventricular systole.

2

7. (continued)

- (c) The diagrams show ECG traces from an individual before and after strenuous exercise.



Calculate the increase in heart rate during exercise.

1

Space for calculation

_____ beats per minute

- (d) Describe how the medulla lowers the heart rate following a period of exercise.

2

[Turn over